[TITLE OF THE INTEGRATED RESEARCH PROJECT]

COMMON INTEGRATED SOLUTION REPORT

Team member

Student ID

Individual component

[Student 1 full name]

[Student 1 ID]

[Component 1 title]

[Student 2 full name]

[Student 2 ID]

[Component 2 title]

[Student 3 full name]

[Student 3 ID]

[Component 3 title]

[Student 4 full name]

[Student 4 ID]

[Component 4 title]

Report submitted in partial fulfilment of the requirements for the

[Official degree title]

[Specialisation, if applicable]

Department of [Department name]

Sri Lanka Institute of Information Technology

Sri Lanka

[Month Year]

DECLARATION AND SUPERVISOR CERTIFICATION

Candidate declaration. We declare that this common report is our own work and does not incorporate, without acknowledgement, material previously submitted for a degree or diploma at any university or institute of higher learning. To the best of our knowledge, it contains no material previously published or written by another person except where acknowledgement is made in the text. We grant the Sri Lanka Institute of Information Technology the non-exclusive right to reproduce and distribute this report, in whole or in part, in print, electronic, or other media, while retaining the right to use the content in future work.

Academic integrity and AI disclosure. We have disclosed all use of generative AI and other automated tools in accordance with the module policy. We remain responsible for the accuracy, originality, citations, code, data, analyses, and conclusions in this report.

Student name

Student ID

Signature

Date

[Student 1]

[ID]

[Student 2]

[ID]

[Student 3]

[ID]

[Student 4]

[ID]

Supervisor certification. The above candidates carried out this research project under my supervision.

Role and name

Signature

Date

Supervisor: [Name]

Co-supervisor: [Name]

ABSTRACT

[Guidance - delete before submission: Maximum 300 words. State the common problem, the four-component integrated solution, overall methodology, principal integrated results, conclusion, and value of the complete solution. Do not combine four separate abstracts.]

[Write the abstract here as one coherent paragraph or a small

number of connected paragraphs.]

Keywords: [three to five keywords, separated by commas]

ACKNOWLEDGEMENTS

[Guidance - delete before submission: Acknowledge supervision, stakeholder access, data or facilities, technical assistance, funding, and other support received by the team. Keep the wording proportionate and professional.]

[Replace this paragraph with the required content.]

TABLE OF CONTENTS

[Guidance - delete before submission: In Microsoft Word, press Ctrl + A and then F9 to update the table of contents, all lists, cross-references, and page numbers before submission.]

Right-click and select Update Field

LIST OF FIGURES

[Guidance - delete before submission: Use Word's Insert Caption command with the label Figure. Number figures by chapter and cite every figure in the text.]

Right-click and select Update Field

LIST OF TABLES

[Guidance - delete before submission: Use Word's Insert Caption command with the label Table. Number tables by chapter and cite every table in the text.]

Right-click and select Update Field

LIST OF ABBREVIATIONS

[Guidance - delete before submission: List abbreviations alphabetically. Define each abbreviation at first use in the main text even when it appears here.]

Abbreviation

Meaning

[ABC]

[Full term]

[ABC]

[Full term]

[ABC]

[Full term]

1 INTRODUCTION AND PROJECT CONTEXT

1.1 Background and context

[Guidance - delete before submission: Introduce the real-world domain, stakeholders, and setting. Establish why the common problem matters using current evidence and citations.]

1.2 Stakeholders and user needs

[Guidance - delete before submission: Identify primary and

secondary stakeholders, user personas where appropriate, and the needs confirmed through stakeholder engagement.]

1.3 Alignment with the Sustainable Development Goals

[Guidance - delete before submission: Identify the relevant UN Sustainable Development Goal or Goals and explain the project's intended contribution. Avoid unsupported claims of impact.]

1.4 Purpose and scope of the common report

[Guidance - delete before submission: Explain that this report presents the integrated problem and solution, while the four individual theses provide the detailed component-level research and development evidence.]

2 OVERALL PROBLEM DEFINITION AND FOUR SUBPROBLEMS

2.1 Overall problem definition

[Guidance - delete before submission: State one precise, evidence-based common problem. Separate the problem from the proposed technology or solution.]

2.2 Root causes and overall research gap

[Guidance - delete before submission: Synthesize the main causes, limitations of current practice, and the overarching gap that requires a four-component solution.]

2.3 Subproblem and component mapping

[Guidance - delete before submission: Describe the four non-overlapping but interdependent subproblems. Show why all four are necessary to address the common problem.]

2.4 Overall aim and group objectives

[Guidance - delete before submission: State one common aim and measurable group objectives. Map each objective to one or more components and to evidence used for evaluation.]

2.5 Scope assumptions and constraints

[Guidance - delete before submission: Define the boundaries of the complete solution, important assumptions, exclusions, operational constraints, and dependencies.]

3 OVERALL SOLUTION STRUCTURE

3.1 Integrated solution overview

[Guidance - delete before submission: Present the solution as one system or framework, not four unrelated products. Explain the central value proposition and end-to-end outcome.]

3.2 High level architecture

[Guidance - delete before submission: Include a readable architecture diagram showing users, shared services, four components, interfaces, data flows, external systems, and deployment boundaries.]

3.3 Component 1 summary

[Guidance - delete before submission: Briefly state the component owner, subproblem, input, core method, output, and contribution to the overall solution. Leave detailed methods to the individual thesis.]

3.4 Component 2 summary

[Guidance - delete before submission: Use the same structure as Component 1 to enable comparison and demonstrate a deliberate division of work.]

3.5 Component 3 summary

[Guidance - delete before submission: Use the same structure as Component 1 to enable comparison and demonstrate a deliberate

division of work.]

3.6 Component 4 summary

[Guidance - delete before submission: Use the same structure as Component 1 to enable comparison and demonstrate a deliberate division of work.]

3.7 End to end user and data flow

[Guidance - delete before submission: Explain how a user or data item moves through the full solution and where the four components interact.]

4 COMPONENT INTEGRATION AND COHESIVE OPERATION

4.1 Integration strategy

[Guidance - delete before submission: Explain the integration approach, sequence, environments, interface ownership, and how integration risks were managed.]

4.2 Interfaces data contracts and shared models

[Guidance - delete before submission: Document APIs, message formats, schemas, shared identifiers, validation rules, versions, and error-handling responsibilities.]

4.3 Shared nonfunctional requirements

[Guidance - delete before submission: Address performance, usability, security, privacy, reliability, maintainability, scalability, accessibility, and any relevant professional standards.]

4.4 Integrated workflow and exception handling

[Guidance - delete before submission: Describe normal end-to-end operation and important failure paths, recovery behaviour, and fallbacks.]

4.5 Team contribution and dependency matrix

[Guidance - delete before submission: Provide a transparent mapping of students to components, research tasks, development tasks, integration work, test evidence, and shared outputs.]

5 INTEGRATED IMPLEMENTATION AND PROJECT EXECUTION

5.1 Common technology stack and infrastructure

[Guidance - delete before submission: Describe shared platforms, repositories, databases, services, environments, and reasons for major common technology choices.]

5.2 Development and integration process

[Guidance - delete before submission: Summarize sprint planning, interface-first work, version control, reviews, continuous integration, issue management, and release practices.]

5.3 Deployment and operational configuration

[Guidance - delete before submission: Explain how the integrated solution is built, configured, deployed, monitored, backed up, and restored.]

5.4 Evidence of team execution

[Guidance - delete before submission: Summarize evidence from MS Planner, Teams, Git history, meeting records, and research diaries. Do not paste raw activity logs into the chapter.]

6 INTEGRATED TESTING

EVALUATION AND VALIDATION

6.1 Evaluation framework and success criteria

[Guidance - delete before submission: Define group-level evaluation questions, acceptance criteria, baselines, metrics, datasets, participants, and analysis methods before presenting results.]

6.2 Integration and end to end testing

[Guidance - delete before submission: Report interface, system, regression, security, performance, usability, and user acceptance tests relevant to the complete solution.]

6.3 Overall results

[Guidance - delete before submission: Present synthesized group-level results with appropriately labelled tables and figures. Point readers to individual theses for detailed component experiments.]

6.4 Stakeholder and user validation

[Guidance - delete before submission: Report who evaluated the complete solution, how evidence was collected, the principal findings, changes made, and remaining concerns.]

6.5 Achievement of common objectives

[Guidance - delete before submission: Map evidence to every group objective and state whether it was fully, partly, or not achieved, with justification.]

6.6 Integrated limitations and threats

[Guidance - delete before submission: Discuss system-level limitations, dependencies between components, validity threats, generalisability, and unresolved integration risks.]

7 COMMERCIALISATION SUSTAINABILITY AND RESPONSIBLE INNOVATION

7.1 Value proposition and beneficiaries

[Guidance - delete before submission: Explain the value created for customers, users, partners, and society, supported by stakeholder evidence.]

7.2 Market alternatives and differentiation

[Guidance - delete before submission: Identify target segments,

competing or substitute solutions, and defensible differentiation of the integrated product.]

7.3 Business model and feasibility

[Guidance - delete before submission: Summarize channels, partners, resources, costs, revenue or sustainability model, and evidence for technical and operational feasibility.]

7.4 Adoption intellectual property and compliance

[Guidance - delete before submission: Discuss adoption barriers, ownership, licensing, data protection, ethics, safety, relevant regulation, and professional responsibilities.]

7.5 Sustainable Development Goal contribution

[Guidance - delete before submission: Evaluate the plausible contribution to the selected SDG targets, possible negative effects, and measures needed to sustain benefits.]

8 CONCLUSION AND FUTURE DIRECTIONS

8.1 Overall conclusion

[Guidance - delete before submission: Answer the common problem directly using the integrated evidence. Do not merely repeat chapter summaries.]

8.2 Principal integrated contribution

[Guidance - delete before submission: State what the team has produced or learned that is valuable beyond the four individual components.]

8.3 Future directions

[Guidance - delete before submission: Prioritize realistic work required for stronger evidence, deployment, adoption, scaling, or continued research.]

REFERENCES

[Guidance - delete before submission: Include only sources cited in this common report. Use one referencing style consistently; IEEE is recommended for IT4010 unless the department instructs otherwise. Single-space each entry and leave one blank line between entries.]

[[1] Replace with the first complete reference entry.]

APPENDICES

[Guidance - delete before submission: Include only material needed to verify the integrated solution, such as interface specifications,

contribution evidence, integrated test matrices, approvals, or extended results. Label each appendix and cite it from the text.]

Appendix A Integrated interface specification

[Guidance - delete before submission: Replace this appendix title and content as appropriate.]

Appendix B Team contribution declaration

[Guidance - delete before submission: Provide a concise, agreed record of individual and shared contributions. All four members should confirm it.]

TEMPLATE CONVENTIONS

[Guidance - delete before submission: Delete this entire section before final submission. It is placed at the end so that the official title page remains the first page while you work.]

• Use the built-in Title, Heading 1, Heading 2, Heading 3, Normal, Caption, and Guidance styles. Do not imitate headings using manual font changes.

• Replace and delete all grey italic guidance text and all square-bracket placeholders before submission.

• Use Times New Roman 12 point for body text and 1.5 line spacing. Abstracts, tables, captions, long quotations, and reference entries may use single spacing.

• Retain A4 page size and the prescribed margins: 40 mm left and bottom; 25 mm top and right.

• Start each chapter on a new page. Use lowercase Roman page numbers for preliminary pages and Arabic page numbers for the main body and appendices.

• Create captions using Word's Insert Caption command. Number figures and tables by chapter and refer to each one by number in the text.

• Before submission, press Ctrl + A and F9 in Microsoft Word to update the table of contents, lists, captions, cross-references, and page numbers.

• Use clear academic language, proofread the final document, disclose AI assistance, and verify every citation, quotation, result, and claim.

• Keep the common report focused on the complete solution, component relationships, integration, overall evaluation, and shared value. Do not paste four individual theses into it.